INDIAN INSTITUTE OF TECHNOLOGY PATNA

EC3102: VLSI DESIGN LAB

Submitted by:

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Roll No: 2301EC56

Group No: TUESDAY GROUP

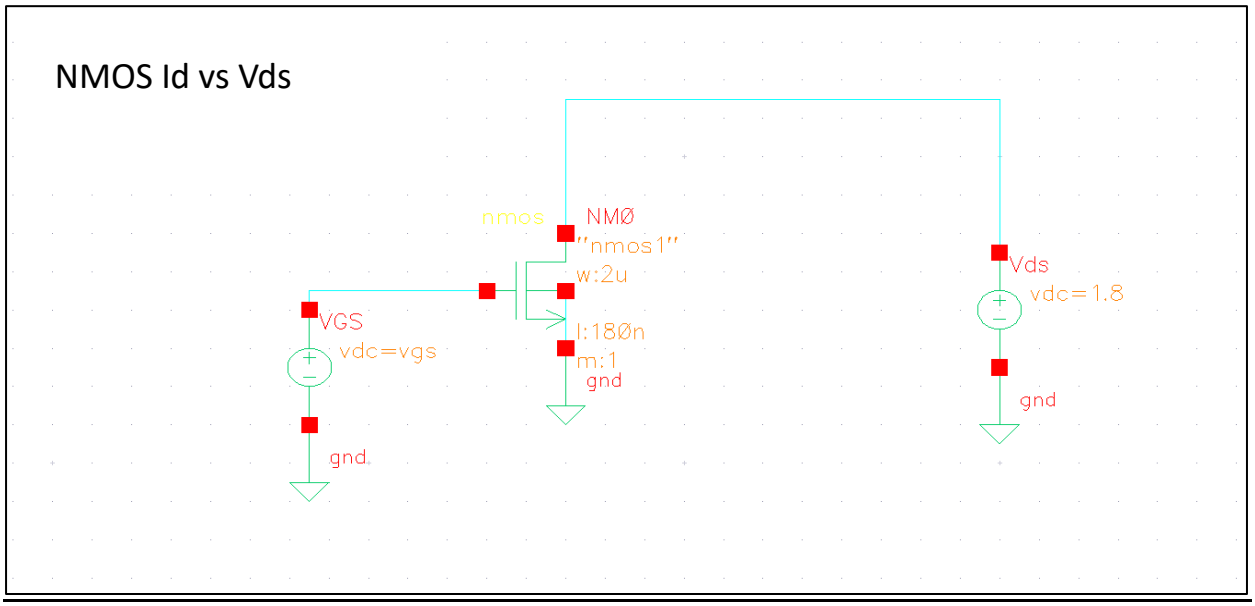
Experiment No: 01

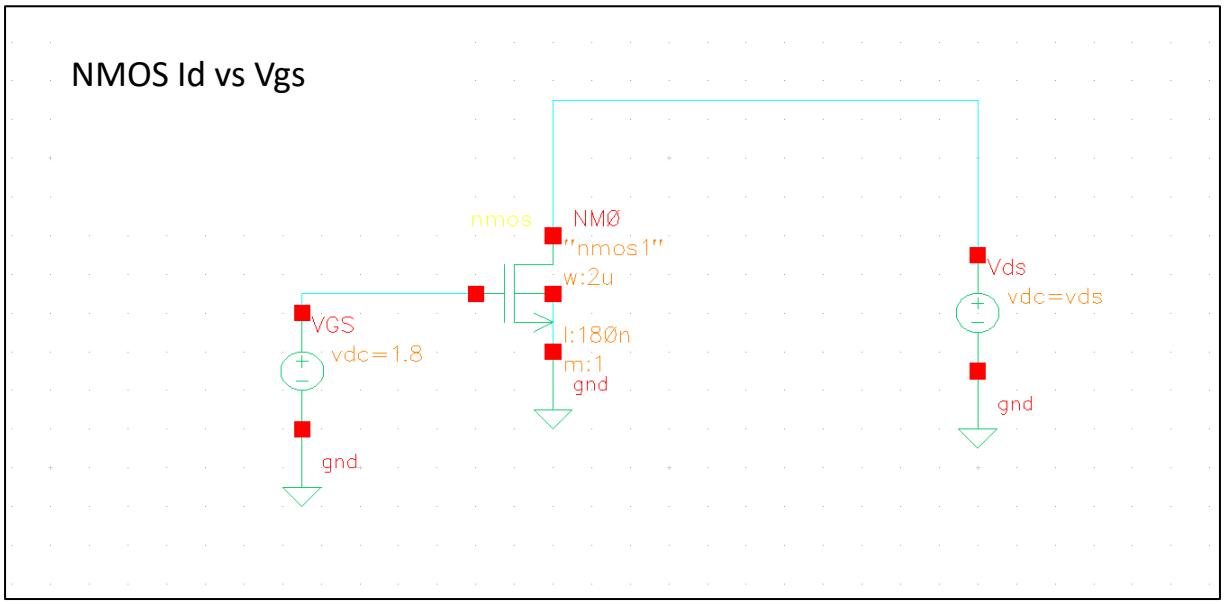
Aim: The aim of this experiment is to study and plot the current-voltage (I-V) characteristics of NMOS and PMOS transistors using Cadence Virtuoso. The experiment involves:

1. Plotting the **output characteristics** (Id vs. Vds) for various constant values of Vgs.
2. Plotting the **transfer characteristics** (Id vs. Vgs) for various constant values of Vds.

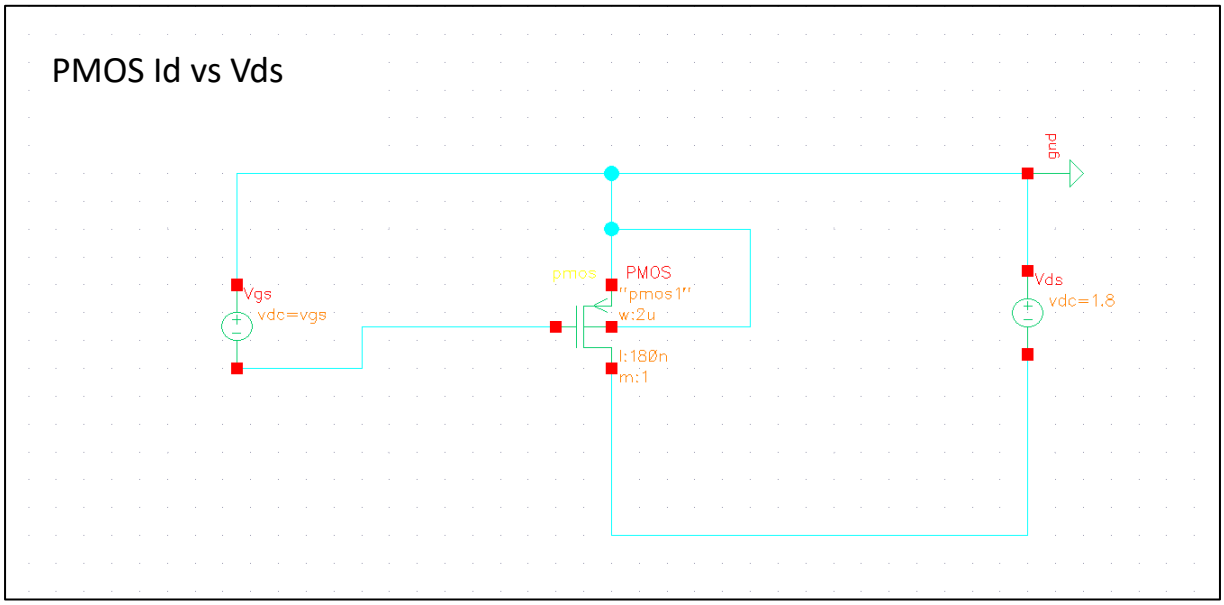
Apparatus: Xming, Putty, Cadence Virtuoso.

Schematics:

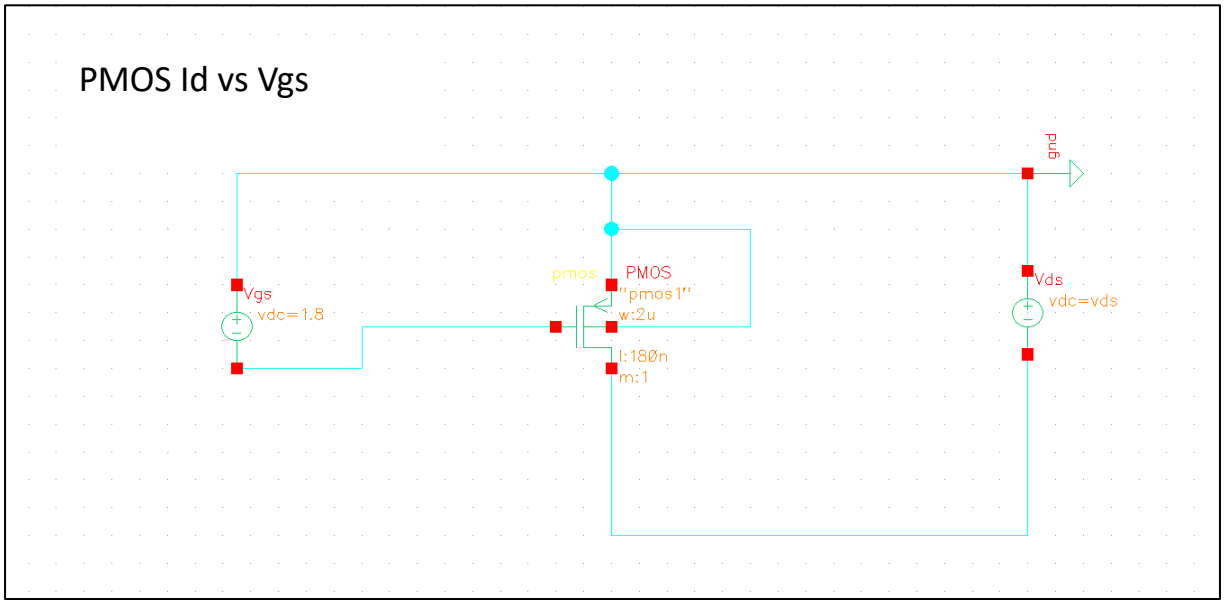




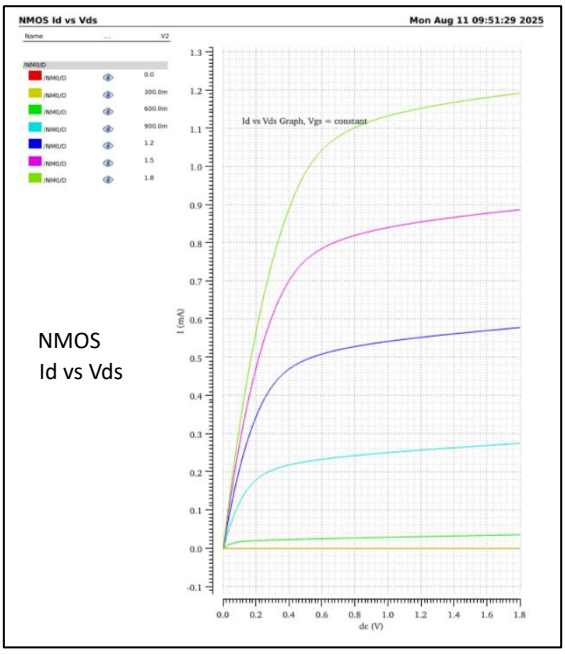
PMOS Id vs Vds



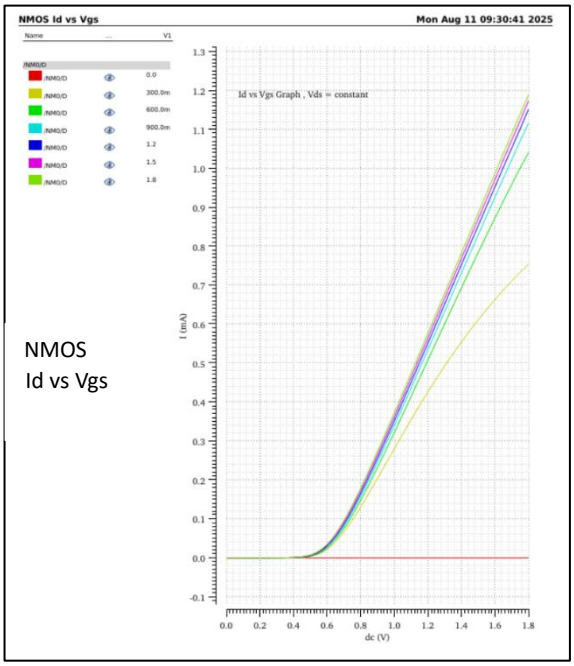
PMOS Id vs Vgs



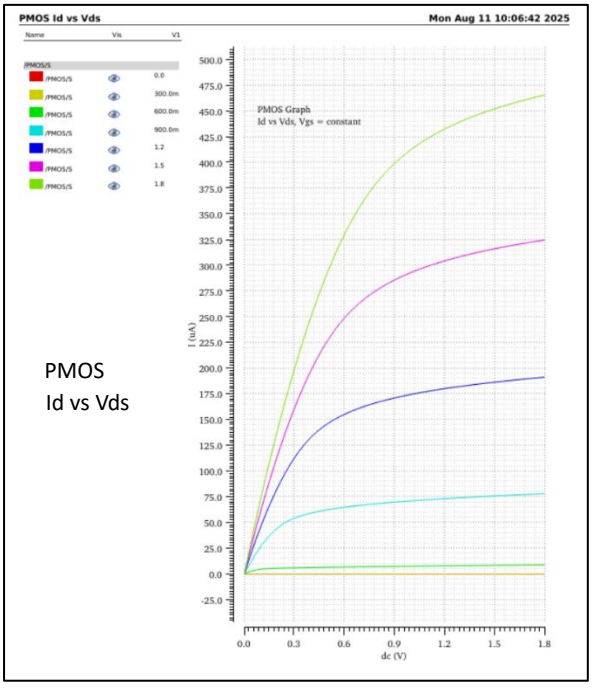
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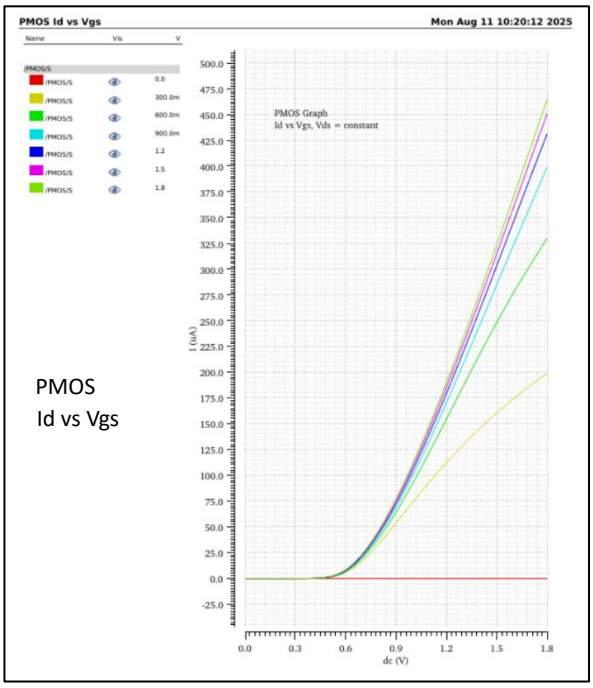
NMOS
Id vs Vds



NMOS
Id vs Vgs



PMOS
Id vs Vds



PMOS
Id vs Vgs

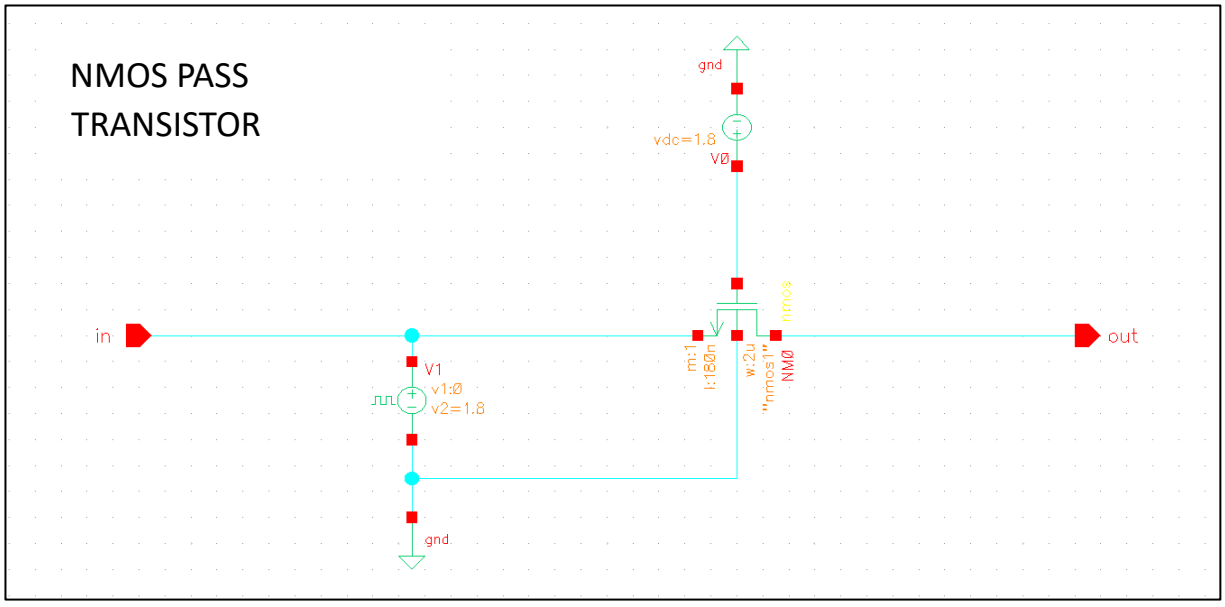
The output and transfer characteristics of NMOS and PMOS are plotted as per the requirements.

Experiment No: 02

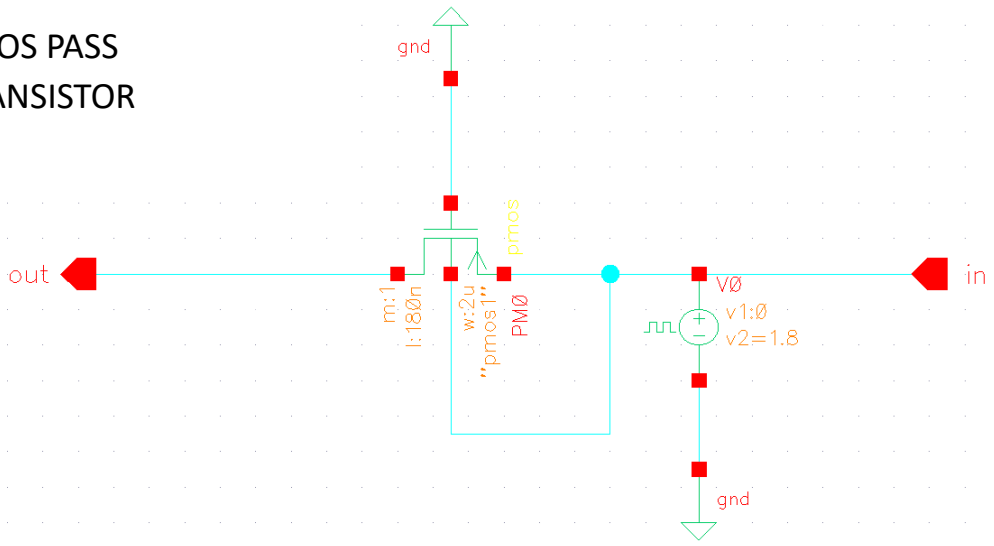
Aim: To design and simulate various logic gates using Pass-Transistor Logic (PTL) and to verify their functionality and limitations.

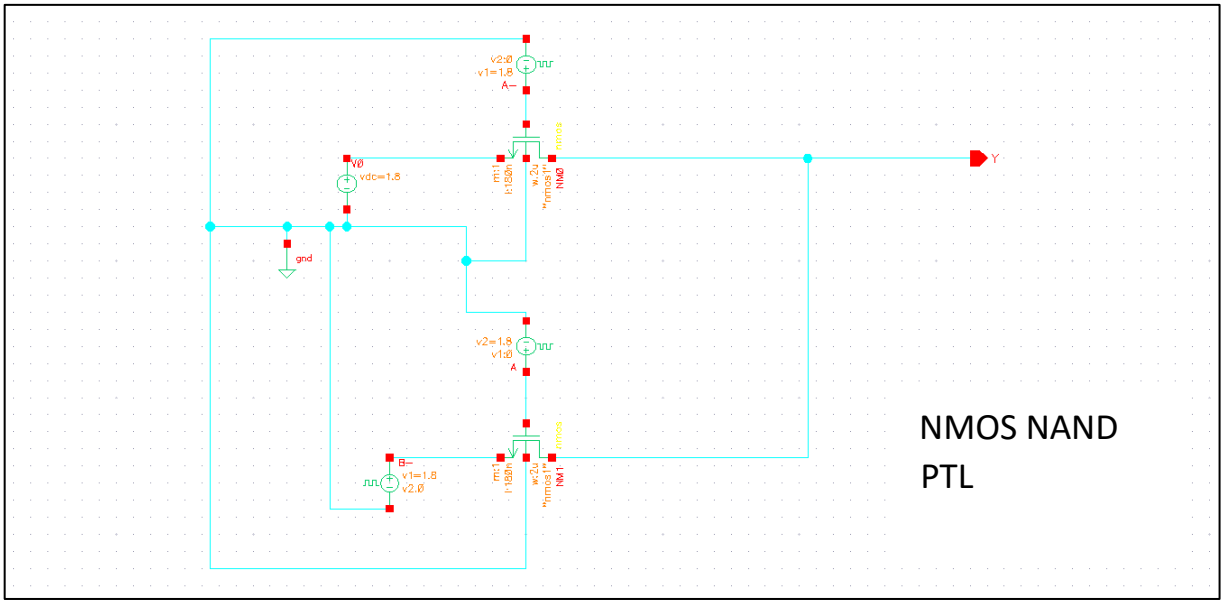
Apparatus: Xming, Putty, Cadence Virtuoso.

Schematics:

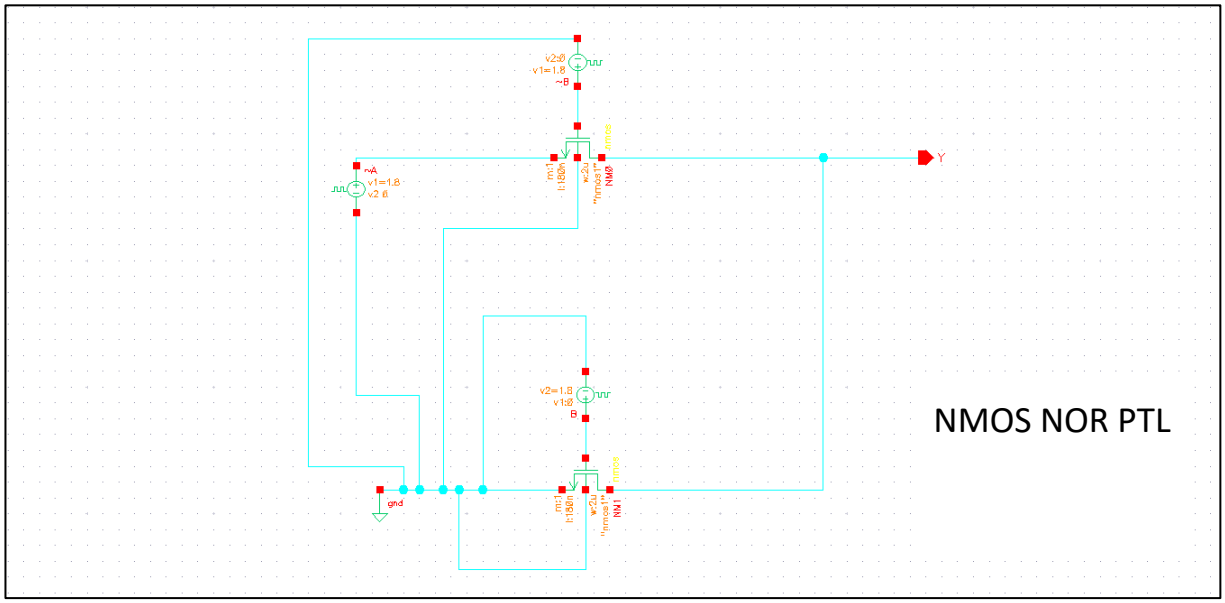


PMOS PASS
TRANSISTOR

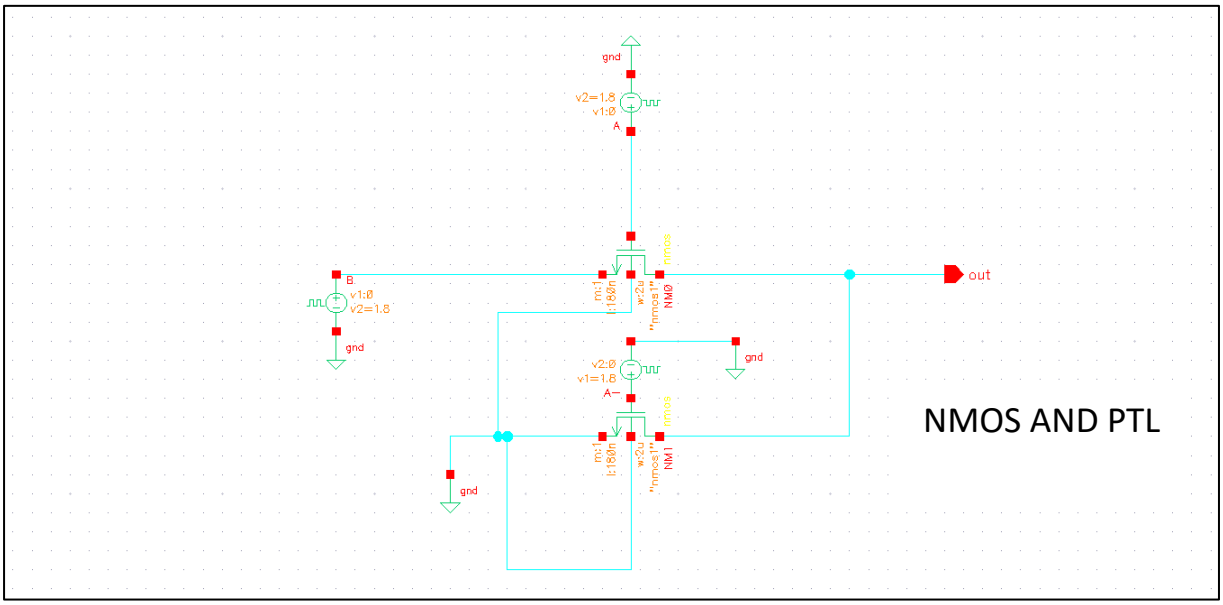




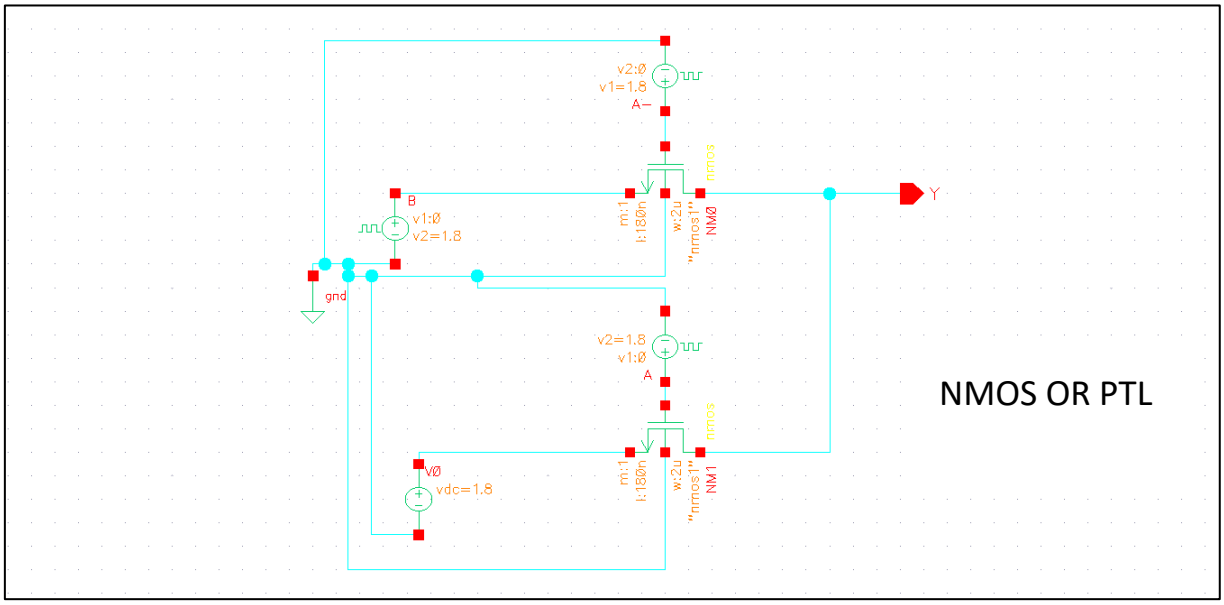
NMOS NAND
PTL



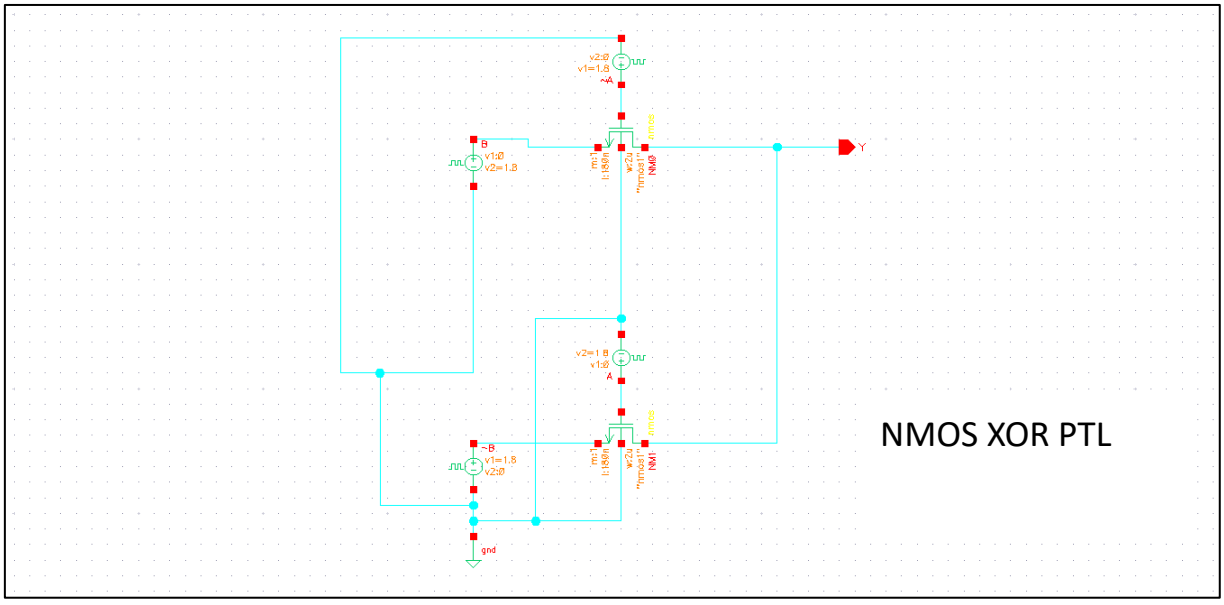
NMOS NOR PTL



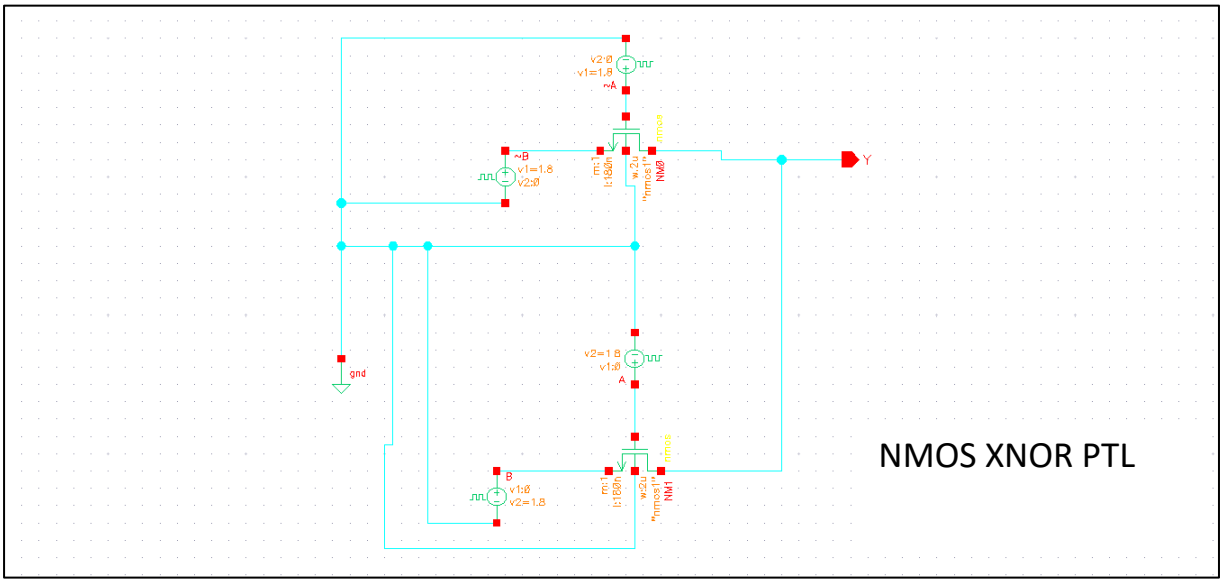
NMOS AND PTL



NMOS OR PTL

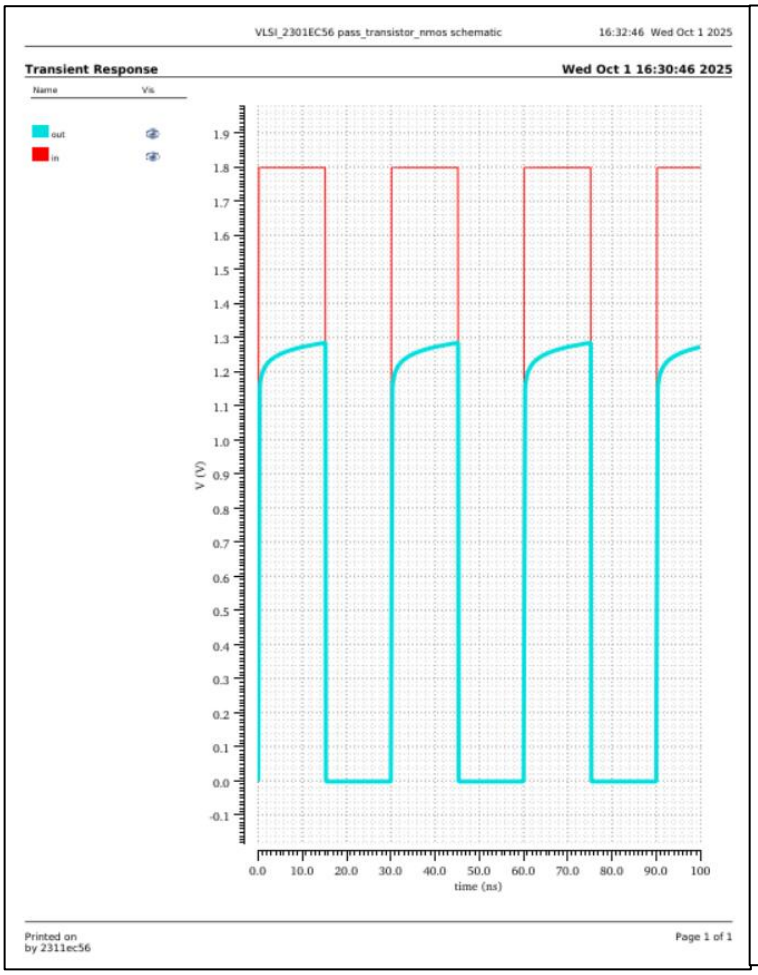


NMOS XOR PTL

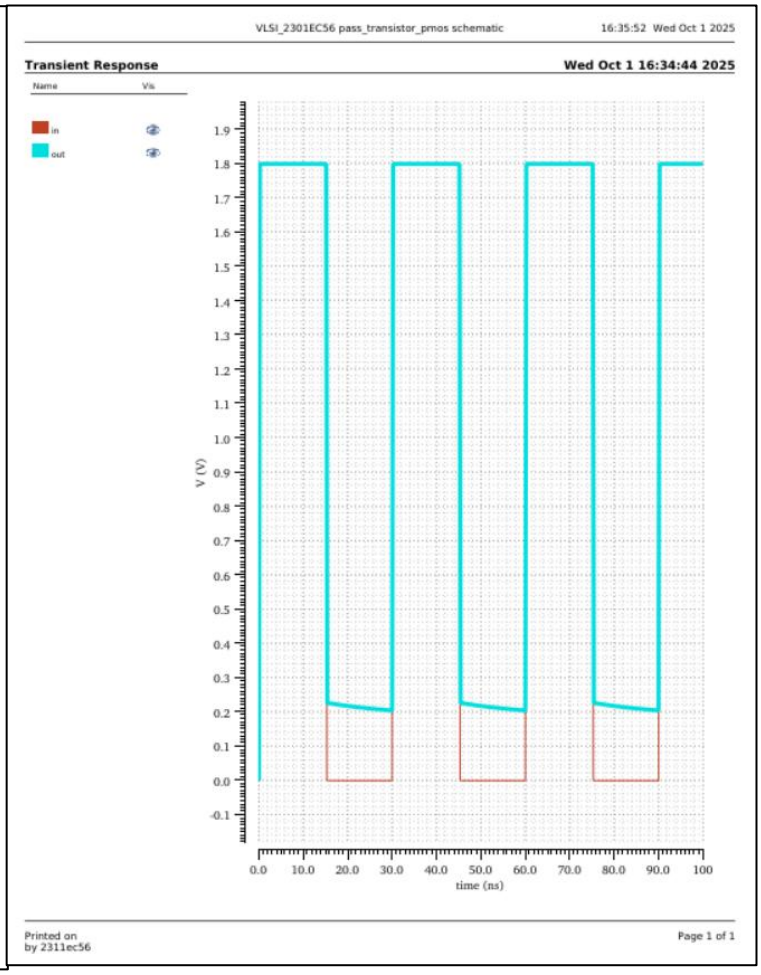


NMOS XNOR PTL

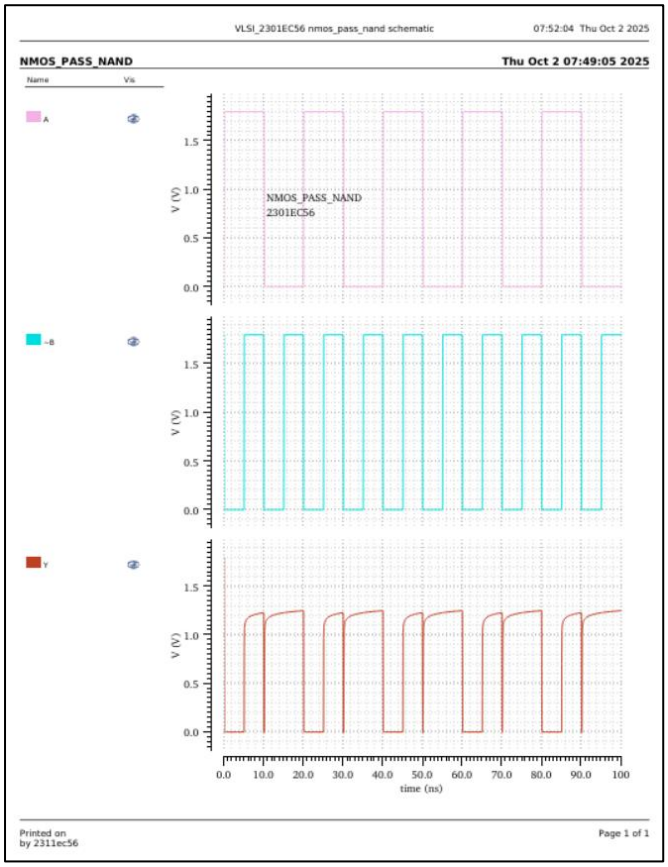
Result:



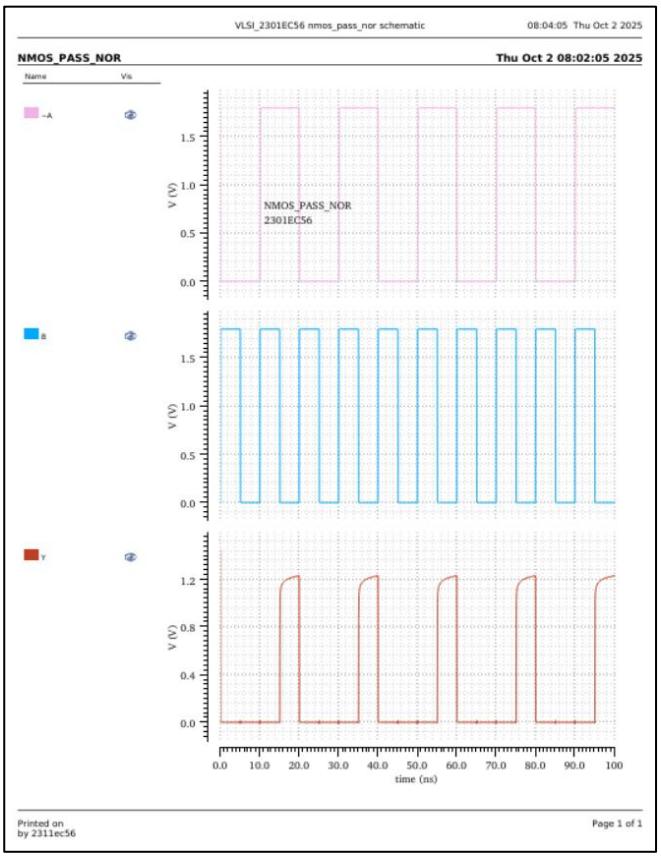
NMOS PASS TRANSISTOR



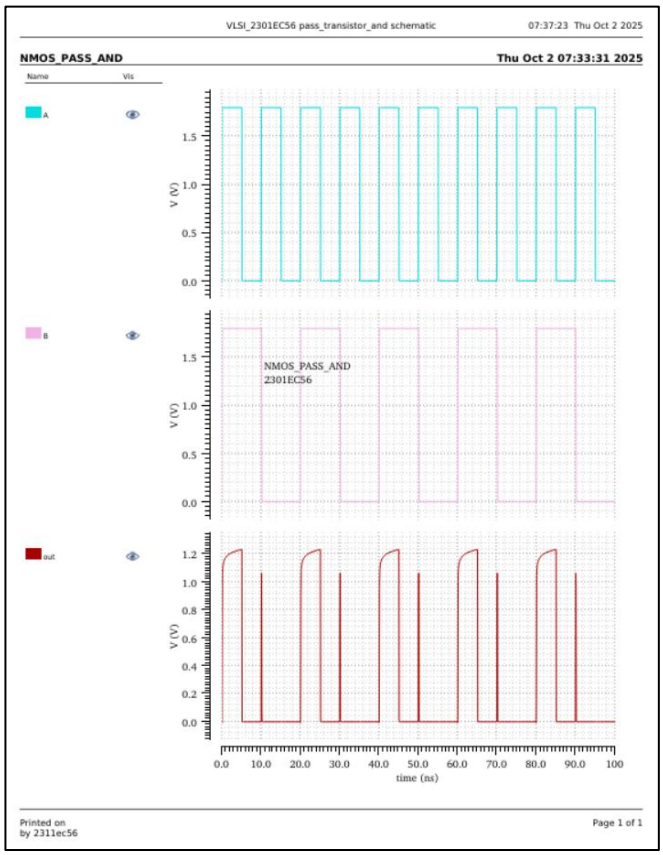
PMOS PASS TRANSISTOR



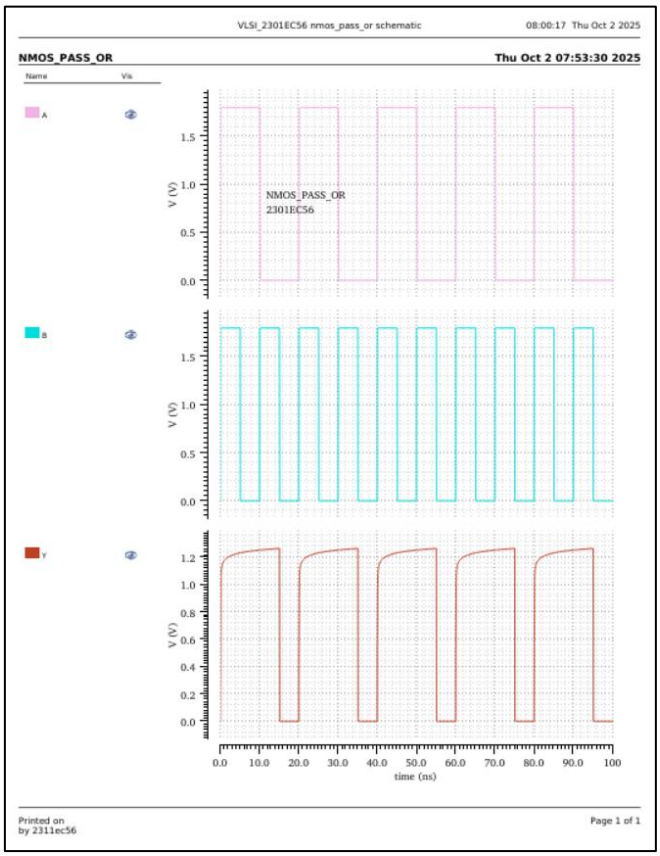
NMOS NAND PTL



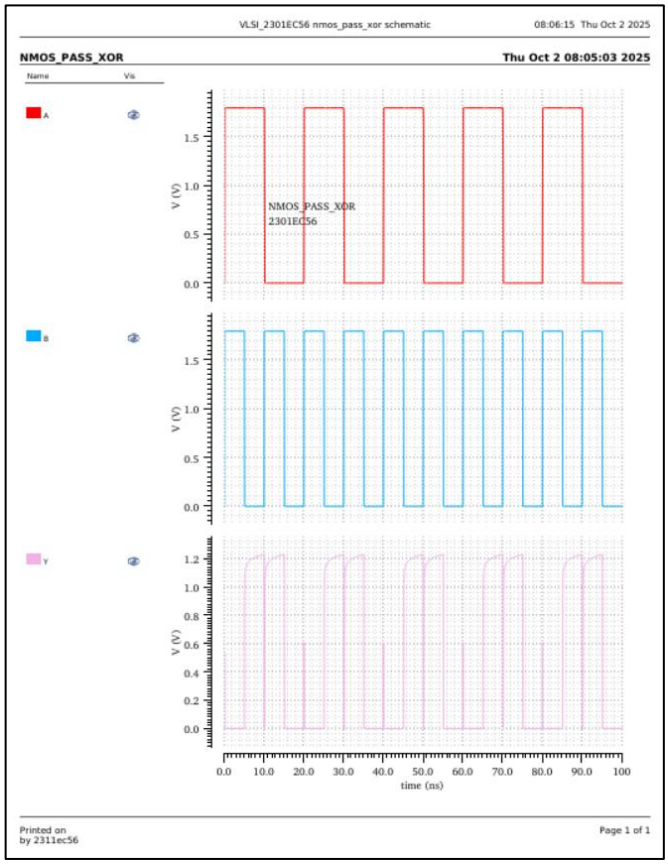
NMOS NOR PTL



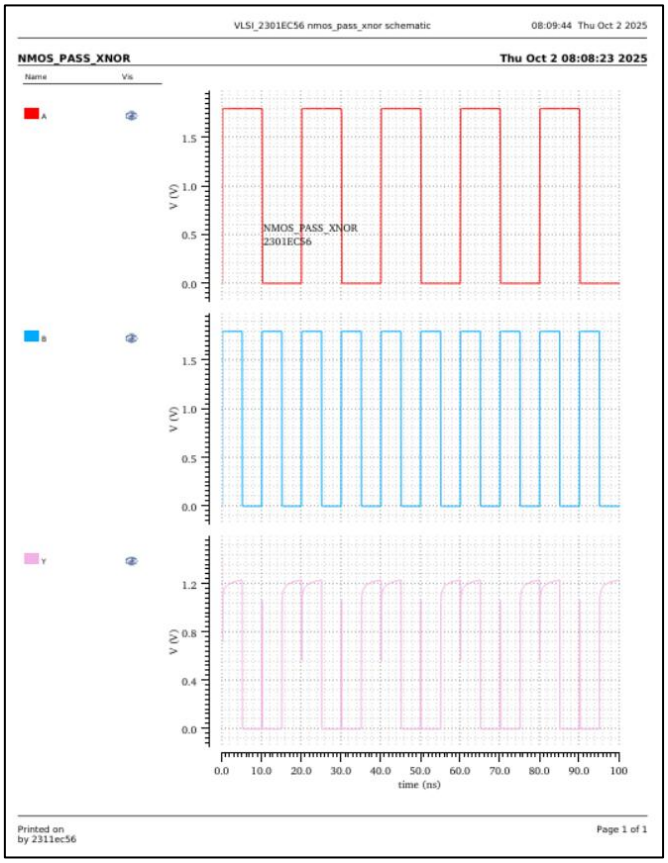
NMOS AND PTL



NMOS OR PTL



NMOS XOR PTL



NMOS XNOR PTL

The graphs of NAND, NOR, AND, OR, XOR, XNOR logic gates are plotted using Pass Transistor Logic (PTL).

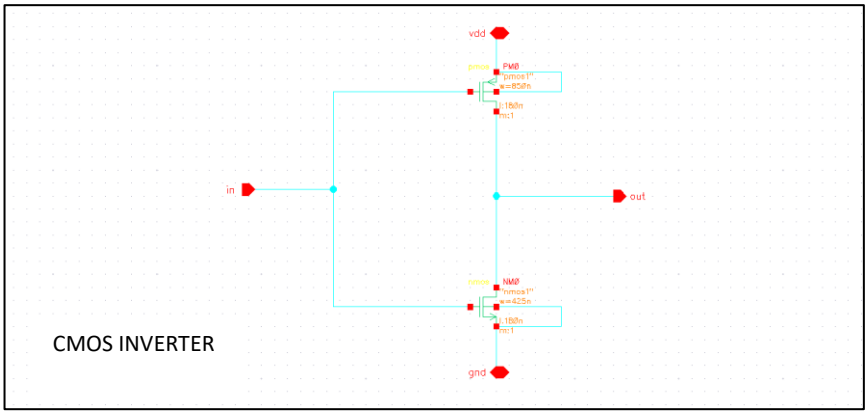
Experiment No: 03

Aim:

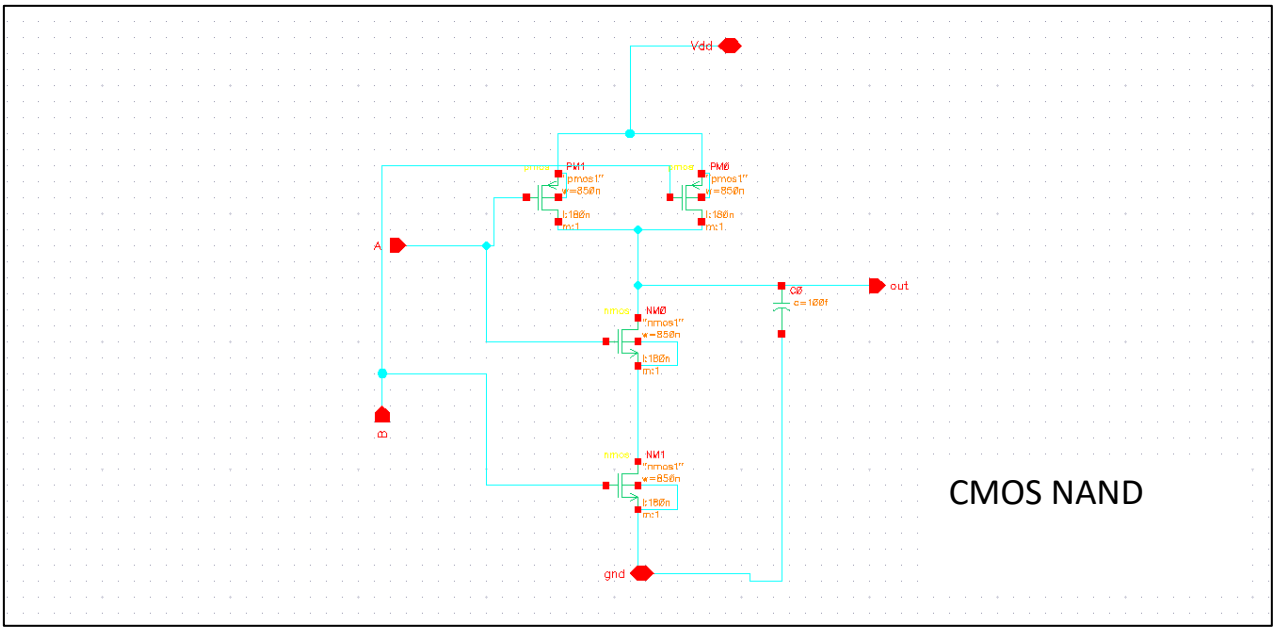
- Design and analyse the universal gates (NAND and NOR) using Cadence VLSI EDA software and CMOS Transistors.
- Perform DC analysis and estimate the noise margins for both configurations.
- Perform transient analysis and estimate the propagation delay for both configurations for different input patterns.
- Identify the worst and best input patterns for both configurations.
- Calculate the maximum switching speed and average power (dynamic and static) if both configurations are operating at 100MHz and 500MHz.

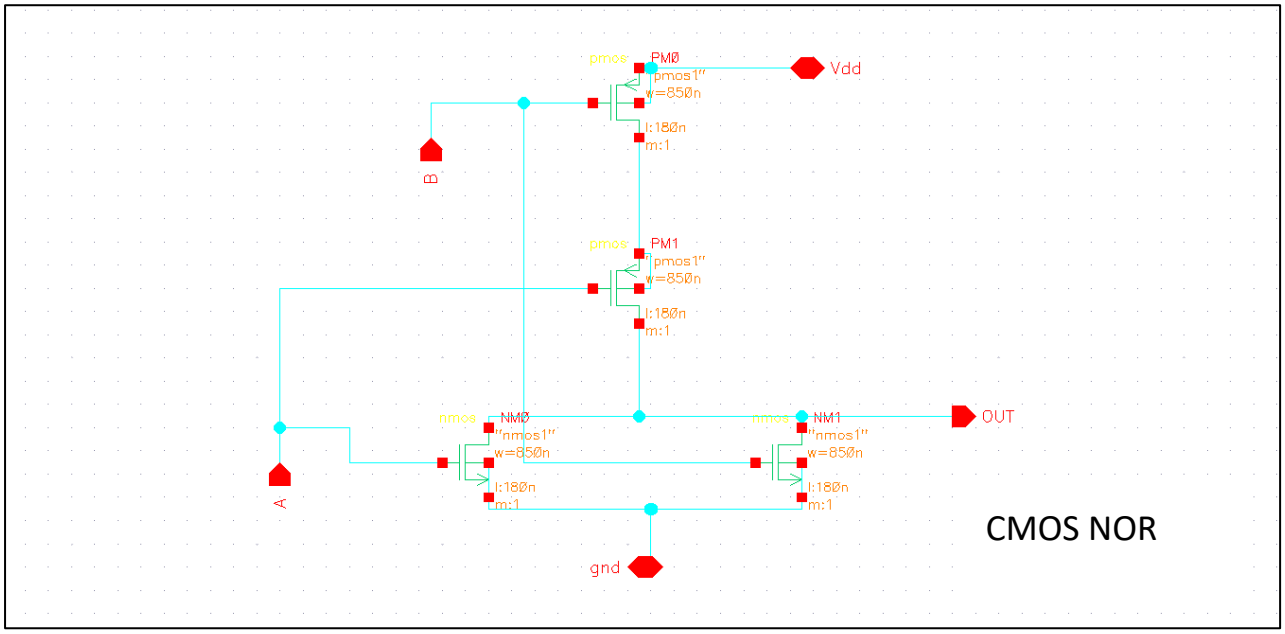
Apparatus: Xming, Putty, Cadence Virtuoso.

Schematics:

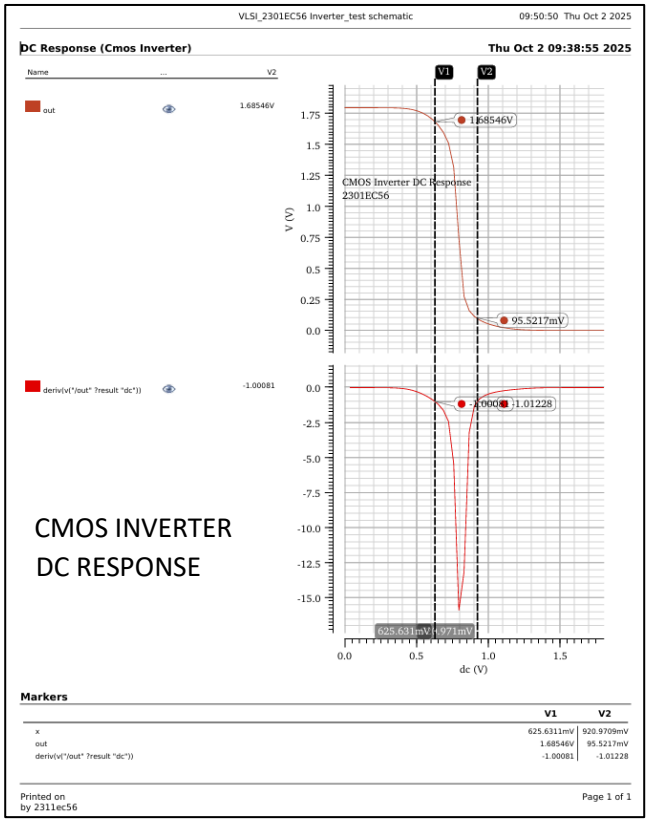


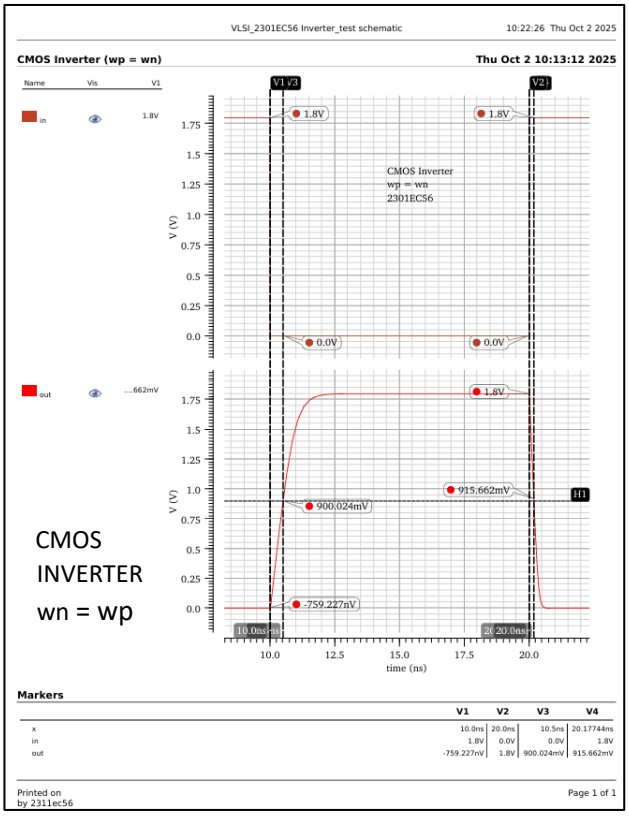
CMOS INVERTER

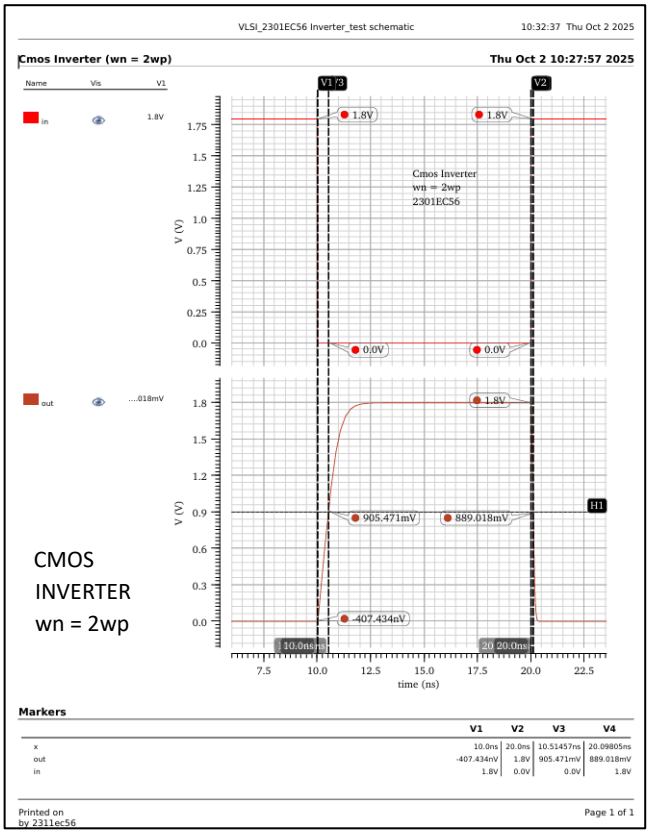

Vdd
PM1
PM0
pmos1"
w=850n
l:180n
m:1
A
B
out
C0
c=120f
NM0
NM1
gnd
CMOS NAND

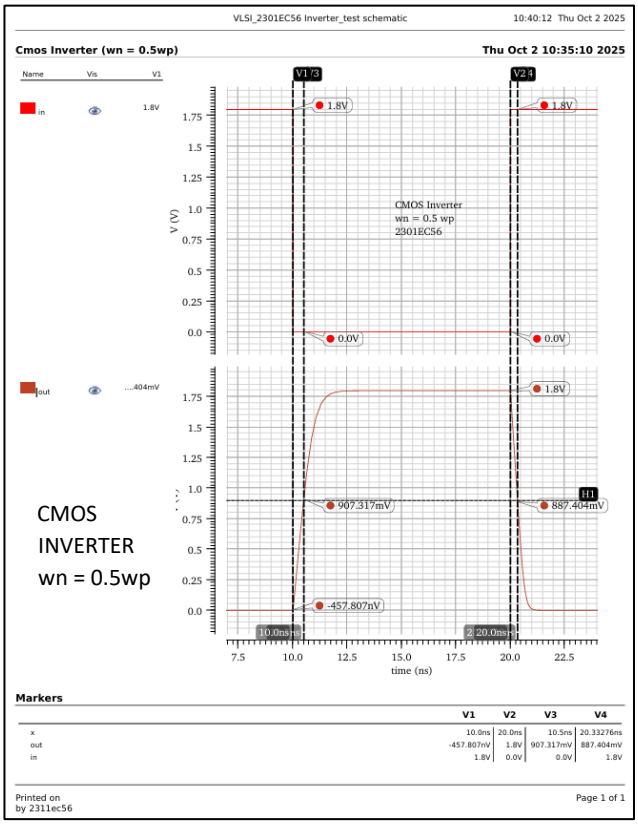

pmos
PM0
Vdd
pmos1"
w=850n
l:180n
m:1
B
PM1
A
nmos
NM0
NM1
OUT
gnd
CMOS NOR

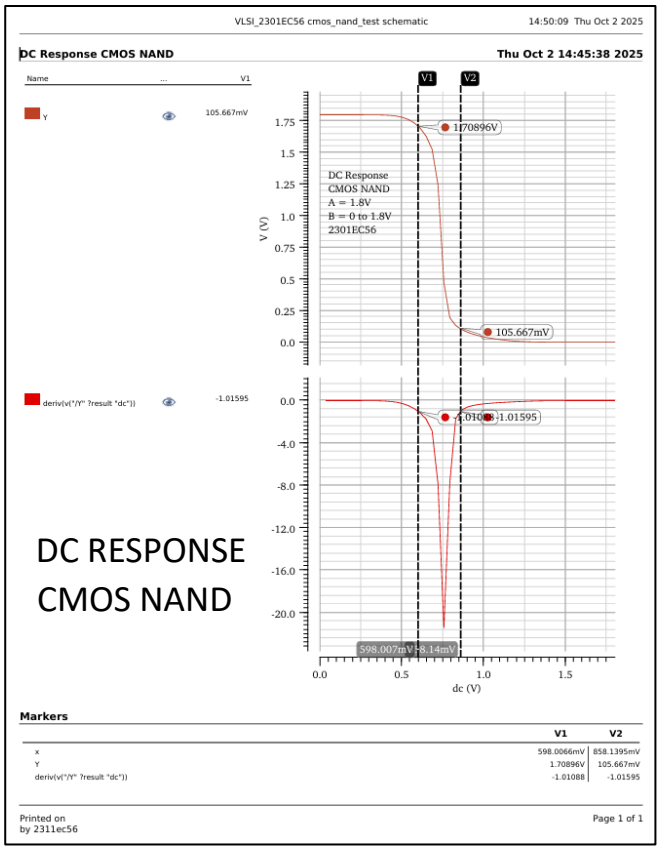
Results:

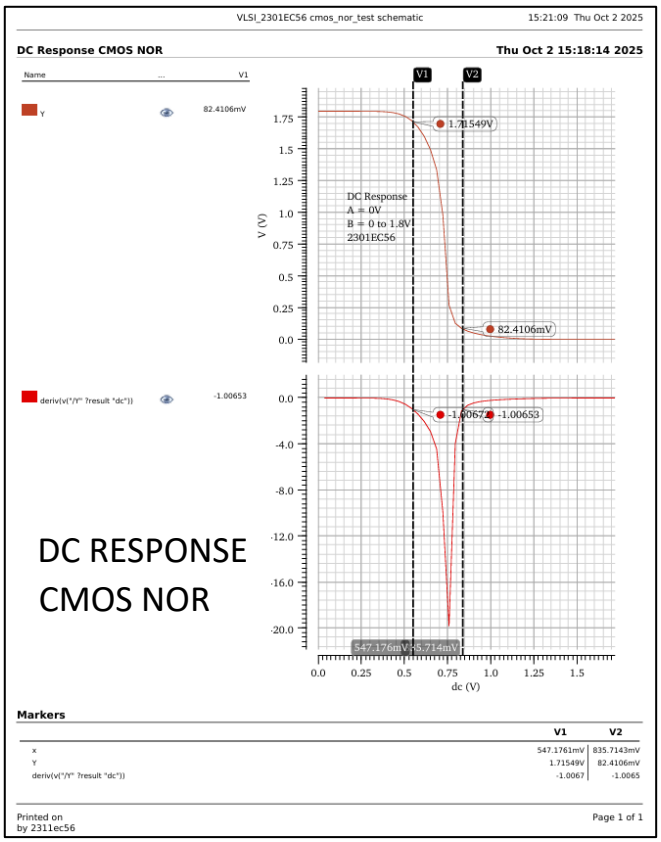

VLSI_2301EC56 Inverter_test schematic
09:50:50 Thu Oct 2 2025
DC Response (Cmos Inverter)
Thu Oct 2 09:38:55 2025
CMOS Inverter DC Response
2301EC56
V (V)
dc (V)
1.68546V
95.5217mV
-1.00081
-1.01228
CMOS INVERTER
DC RESPONSE
Markers
Printed on
by 2311ec56
Page 1 of 1

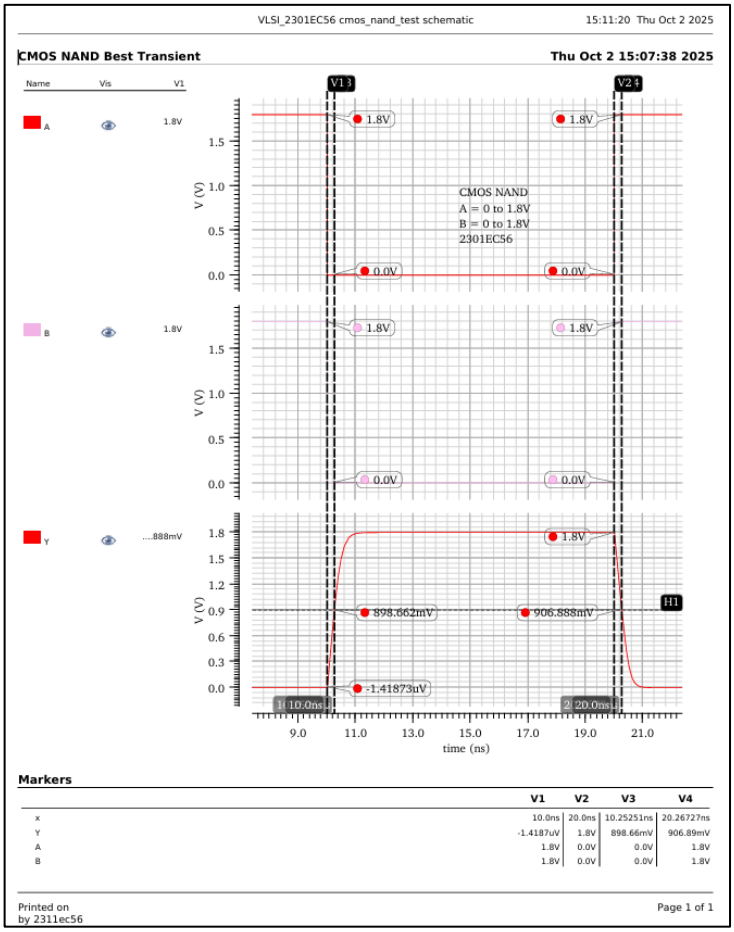

VLSI_2301EC56 Inverter_test schematic
10:22:26 Thu Oct 2 2025
CMOS Inverter (wp = wn)
Thu Oct 2 10:13:12 2025
CMOS Inverter
wp = wn
2301EC56
V (V)
time (ns)
1.8V
0.0V
915.662mV
900.024mV
-759.227nV
CMOS
INVERTER
wn = wp
Markers
Printed on
by 2311ec56
Page 1 of 1


VLSI_2301EC56 Inverter_test schematic
10:32:37 Thu Oct 2 2025
Cmos Inverter (wn = 2wp)
Thu Oct 2 10:27:57 2025
Cmos Inverter
wn = 2wp
2301EC56
V (V)
time (ns)
1.8V
0.0V
905.471mV
889.018mV
-407.434mV
10.0ns
20.0ns
CMOS
INVERTER
wn = 2wp
Markers
V1 V2 V3 V4
x 10.0ns 20.0ns 10.51457ns 20.09805ns
out -407.434mV 1.8V 905.471mV 889.018mV
in 1.8V 0.0V 0.0V 1.8V
Printed on
by 2311ec56
Page 1 of 1

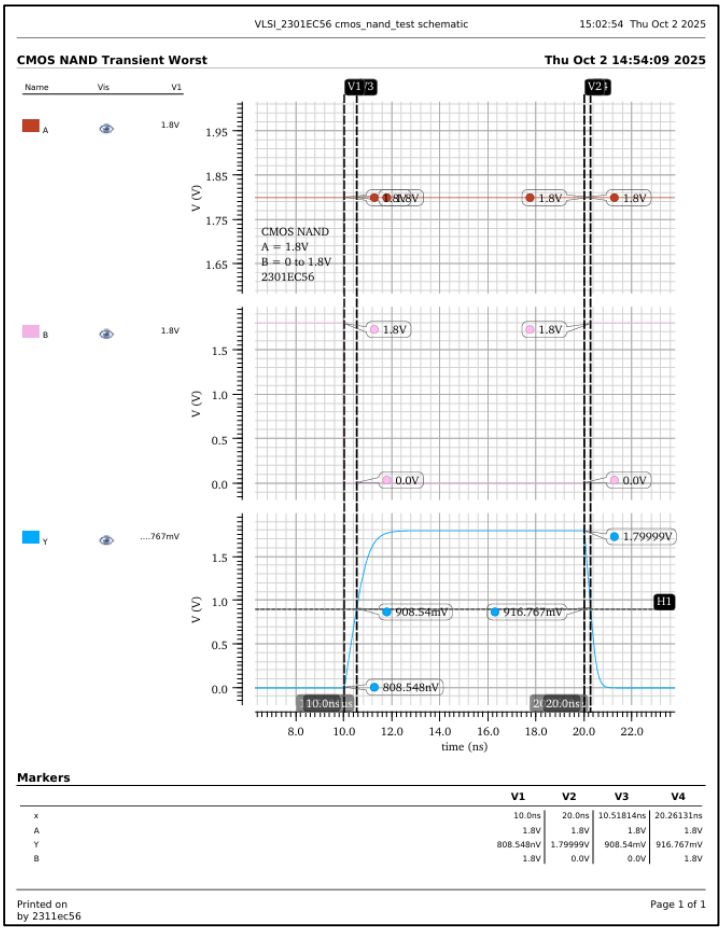

VLSI_2301EC56 Inverter_test schematic
10:40:12 Thu Oct 2 2025
Cmos Inverter (wn = 0.5wp)
Thu Oct 2 10:35:10 2025
CMOS Inverter
wn = 0.5 wp
2301EC56
V (V)
time (ns)
1.8V
0.0V
907.317mV
887.404mV
-457.807mV
10.0ns
20.0ns
CMOS
INVERTER
wn = 0.5wp
Markers
V1 V2 V3 V4
x 10.0ns 20.0ns 10.5ns 20.33276ns
out -457.807mV 1.8V 907.317mV 887.404mV
in 1.8V 0.0V 0.0V 1.8V
Printed on
by 2311ec56
Page 1 of 1


VLSI_2301EC56 cmos_nand_test schematic
14:50:09 Thu Oct 2 2025
DC Response CMOS NAND
Thu Oct 2 14:45:38 2025
DC Response
CMOS NAND
A = 1.8V
B = 0 to 1.8V
2301EC56
V (V)
dc (V)
1.70896V
105.667mV
-1.01088
-1.01595
598.007mV
858.14mV
DC RESPONSE
CMOS NAND
Markers
V1 V2
x 598.0066mV 858.1395mV
Y 1.70896V 105.667mV
deriv(v("/Y" ?result "dc")) -1.01088 -1.01595
Printed on
by 2311ec56
Page 1 of 1

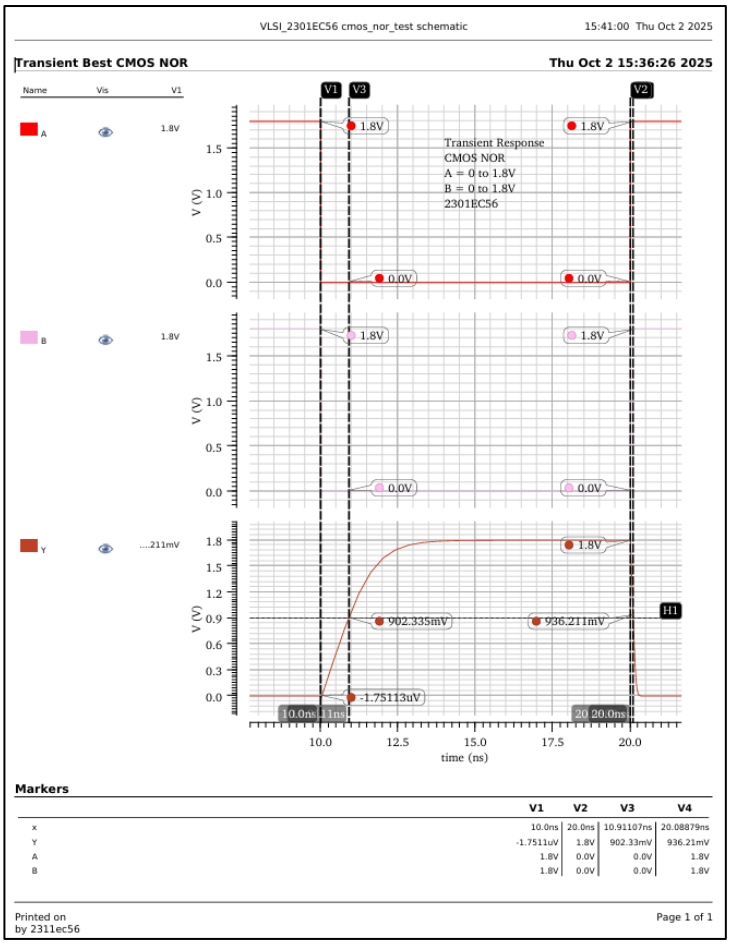

VLSI_2301EC56 cmos_nor_test schematic
15:21:09 Thu Oct 2 2025
DC Response CMOS NOR
Thu Oct 2 15:18:14 2025
DC Response
A = 0V
B = 0 to 1.8V
2301EC56
V (V)
dc (V)
1.71549V
82.4106mV
-1.0067
-1.00653
547.176mV
835.714mV
DC RESPONSE
CMOS NOR
Markers
V1 V2
x 547.1761mV 835.7143mV
Y 1.71549V 82.4106mV
deriv(v("/Y" ?result "dc")) -1.0067 -1.0065
Printed on
by 2311ec56
Page 1 of 1



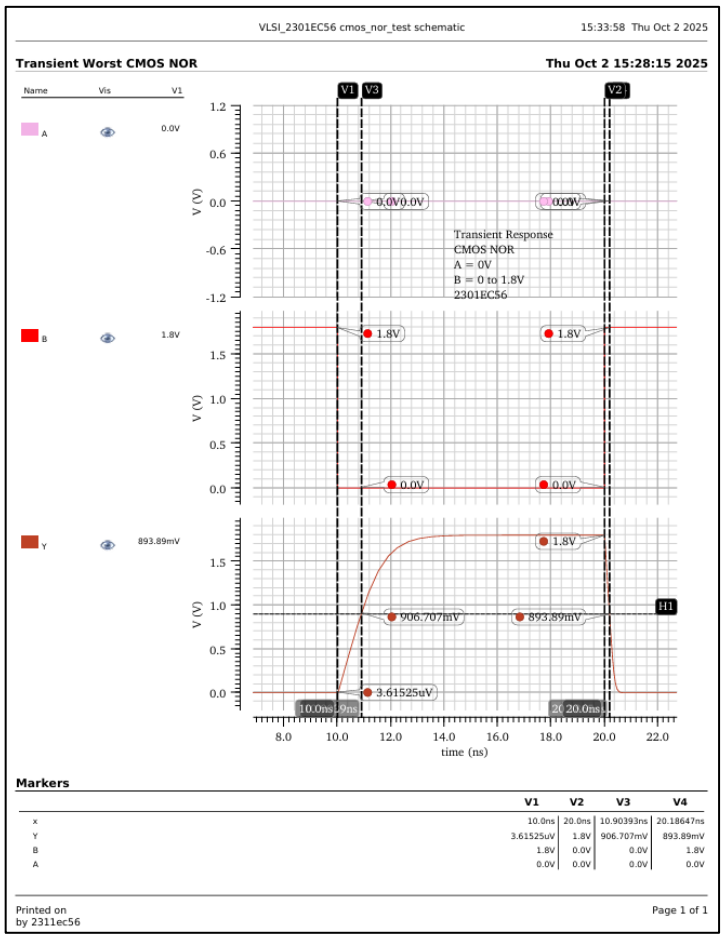
TRANSIENT RESPONSE BEST CMOS NAND



TRANSIENT RESPONSE WORST CMOS NAND



TRANSIENT RESPONSE BEST CMOS NOR



TRANSIENT RESPONSE WORST CMOS NOR

a) DC Analysis for noise margins:

- CMOS INVERTER:
  $NM_L = VI_L – VO_L$ = 625.6311mV – 95.5217mV = 530.1094mV
  $NM_H = VO_H – VI_H$ = 1.68546V – 920.9709mV = 764.4891mV
- CMOS NAND:
  $NM_L = VI_L – VO_L$ = 598.0066mV -- 105.667mV = 492.3396mV
  $NM_H = VO_H – VI_H$ = 1.70896V -- 858.1395mV = 850.8205mV
- CMOS NOR:
  $NM_L = VI_L – VO_L$ = 547.1761mV -- 82.4106mV = 464.7655mV
  $NM_H = VO_H – VI_H$ = 1.71549V -- 835.7143mV = 879.7757mV

b) Transient Analysis and Propagation Delay:

- CMOS INVERTER (wn = wp):
  T_{plh} = 10.5ns – 10.0ns = 0.5ns
  T_{phl} = 20.17744ns – 20.0ns = 0.1744ns

- CMOS INVERTER (wn = 2wp):
  T_{plh} = 10.51457ns– 10.0ns = 0.51457ns
  T_{phl} = 20.09805ns– 20.0ns = 0.09805ns

- CMOS INVERTER (wn = 0.5wp):
  T_{plh} = 10.5ns– 10.0ns = 0.5ns
  T_{phl} = 20.33276ns– 20.0ns = 0.33276ns

- CMOS NAND (Best):
  T_{plh} = 10.25251ns – 10.0ns = 0.25251ns (less propagation delay from low to high)
  T_{phl} = 20.26727ns– 20.0ns = 0.26727ns

- CMOS NAND (Worst):
  T_{plh} = 10.51814ns – 10.0ns = 0.51814ns (high propagation delay from low to high)
  T_{phl} = 20.26131ns – 20.0ns = 0.26131ns

- CMOS NOR (Best):
  T_{plh} = 10.91107ns – 10.0ns = 0.91107ns
  T_{phl} = 20.08879ns – 20.0ns = 0.08879ns (less propagation delay from high to low)

- CMOS NOR (Worst):
  T_{plh} = 10.90393ns– 10.0ns = 0.90393ns
  T_{phl} = 20.18647ns– 20.0ns = 0.18647ns (high propagation delay from high to low)

c) Best and Worst Input Pattern:

For CMOS NAND: Best input pattern is when both inputs are going from high (1.8V) to low (0V), it creates less propagation delay from low to high.

For CMOS NAND: Worst input pattern is when one input is steady (1.8V) and other input is going from high (1.8V) to low (0V), it creates more propagation delay from low to high.

For CMOS NOR: Best input pattern is when both inputs are going from low (0V) to high (1.8V), it creates less propagation delay from high to low.

For CMOS NOR: Worst input pattern is when one input is steady (0V) and other input is going from low (0V) to high (1.8V), it creates more propagation delay from high to low.

d) Maximum Switching Speed and Average Power:

SWITCHING SPEED:

- CMOS INVERTER:
  $F_{MAX} = 1 / (T_{PHL} + T_{PLH}) = 1 / 0.83276 = 1.2008$ GHz
- CMOS NAND:
  $F_{MAX} = 1 / (T_{PHL} + T_{PLH}) = 1 / 0.77945 = 1.2829$ GHz
- CMOS NOR:
  $F_{MAX} = 1 / (T_{PHL} + T_{PLH}) = 1 / 1.0904 = 0.9171$ GHz

AVERAGE POWER:
CMOS NAND (100 MHz) = $35.14 * 10^{-6}$
CMOS NAND (500 MHz) = $147.9 * 10^{-6}$
CMOS NOR (100 MHz) = $35.01 * 10^{-6}$
CMOS NOR (500 MHz) = $98.83 * 10^{-6}$

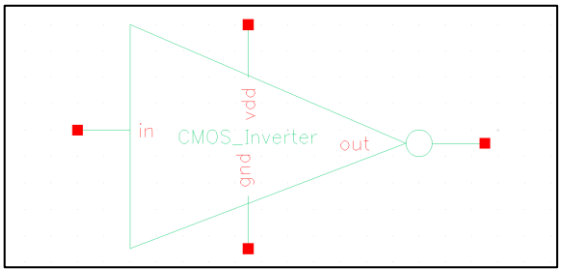
Experiment No: 04

Aim:

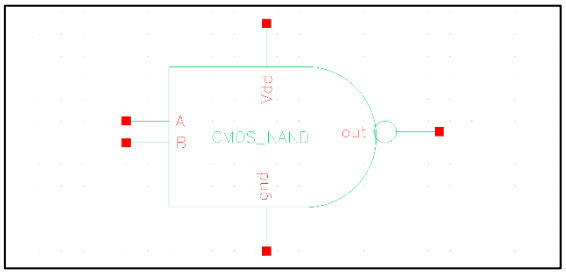
- Create symbols for NAND, NOR, AND, OR, XOR, XNOR and use them in BOB's designs.
- Simulate both designs, using the cell libraries developed in this lab with nominal transistor sizing.
- Identify which design you will choose and why? Justify your choice with experimental data and observations.
- Estimate the switching speed and power in both configurations.

Apparatus: Xming, Putty, Cadence Virtuoso.

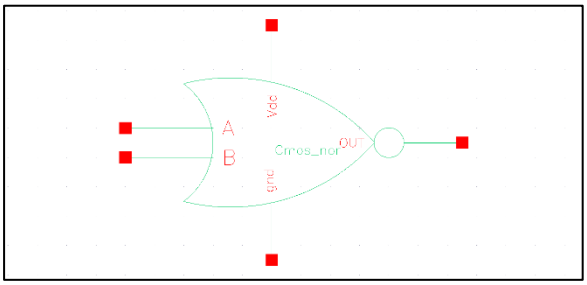
Schematics:



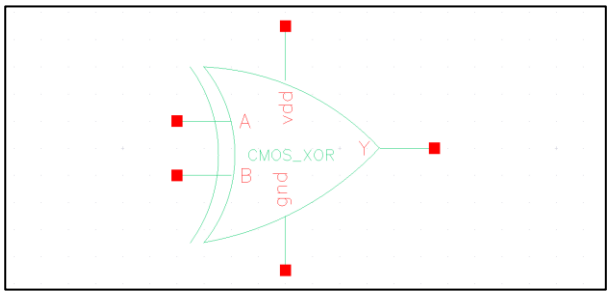
INVERTER



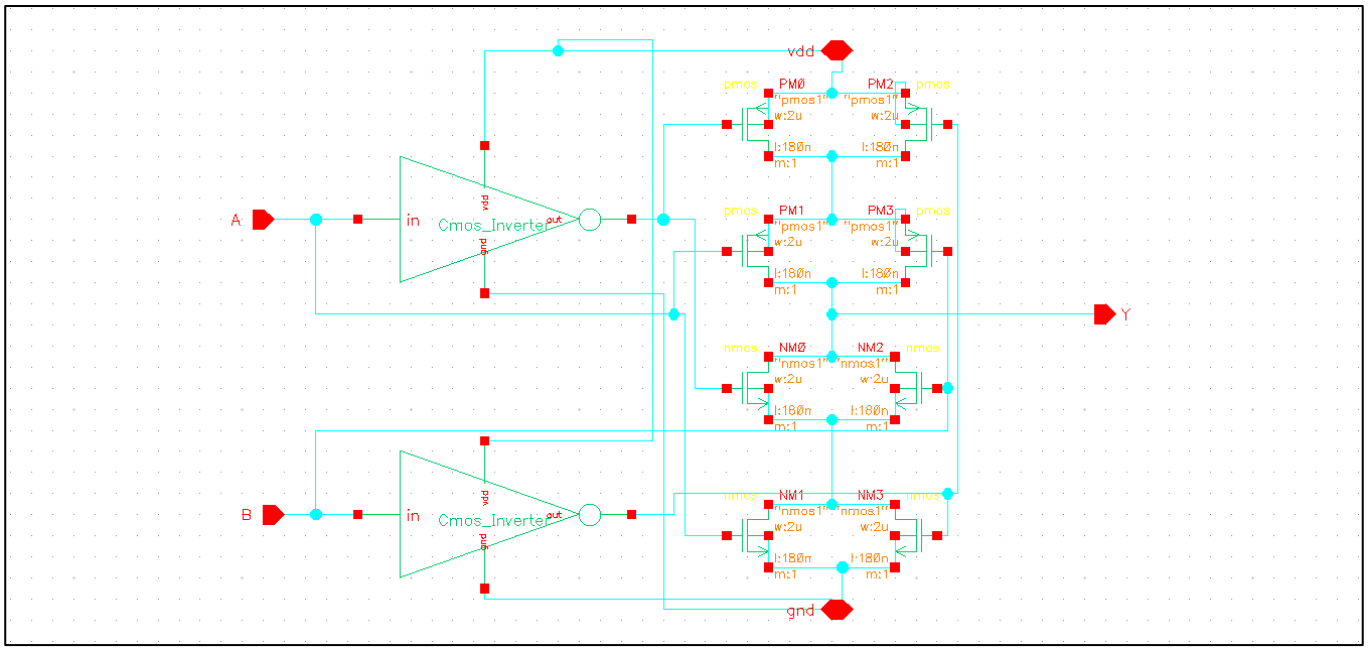
NAND



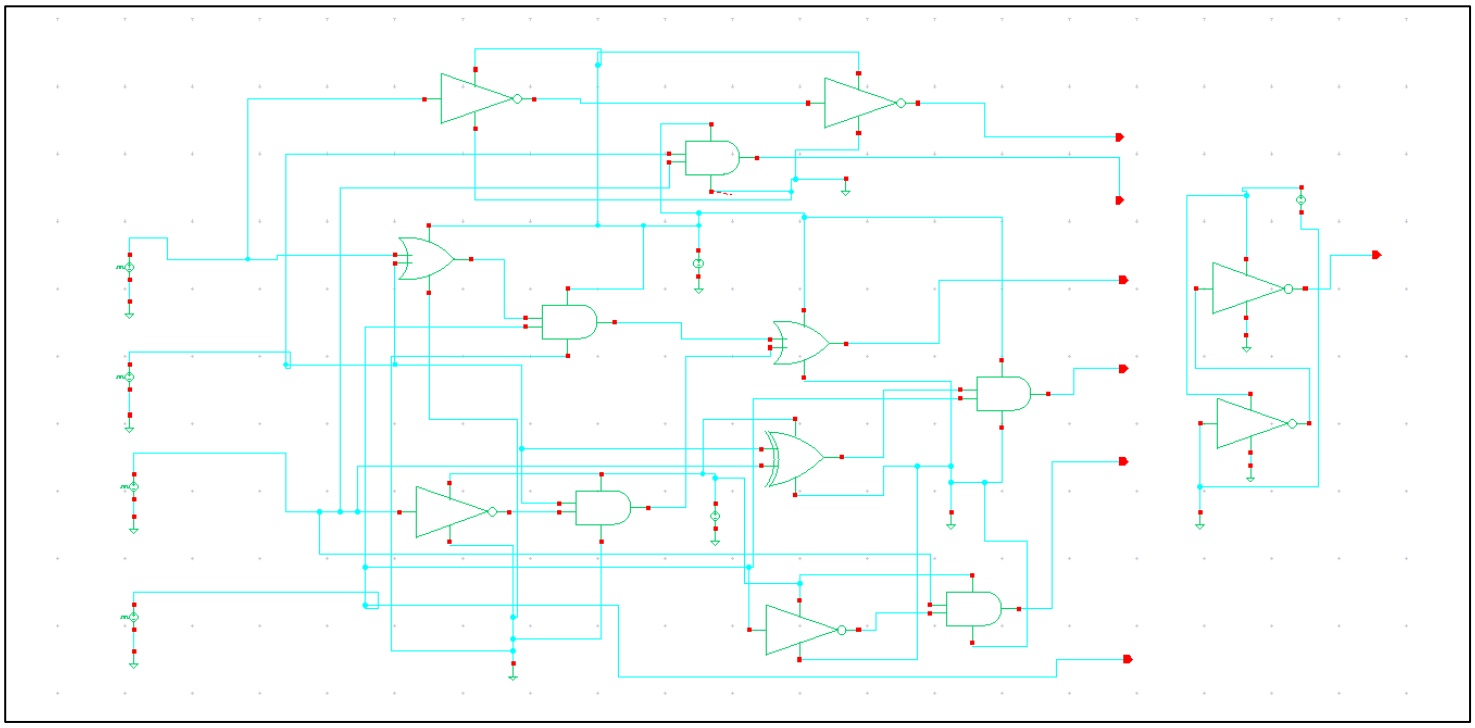
NOR



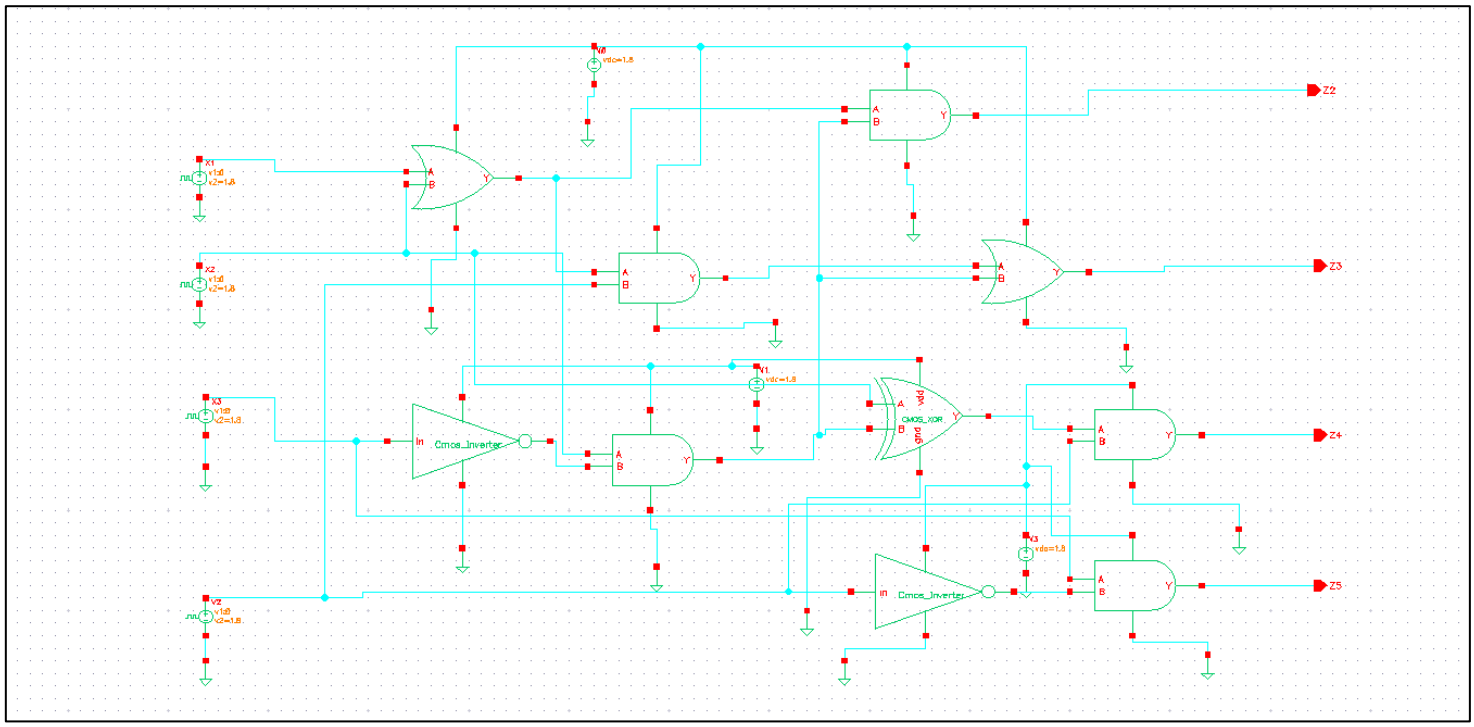
XOR



CMOS XOR CIRCUIT DIAGRAM

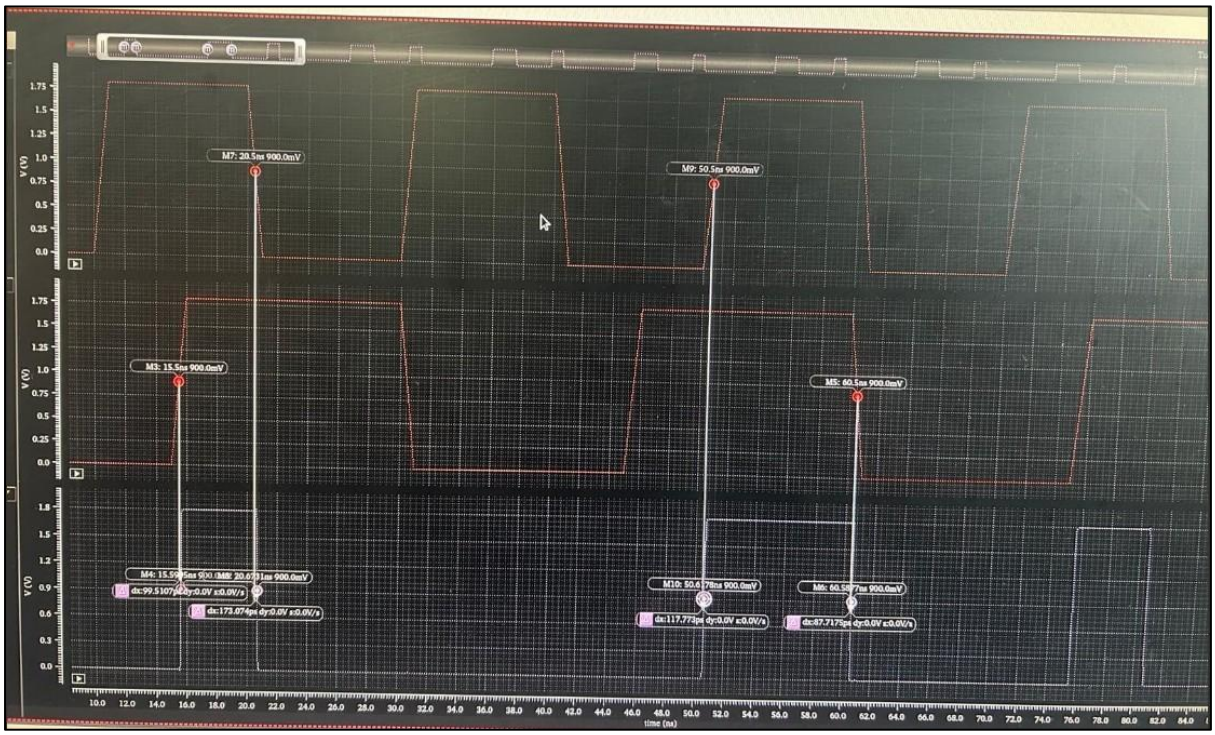


BOB DESIGN 1

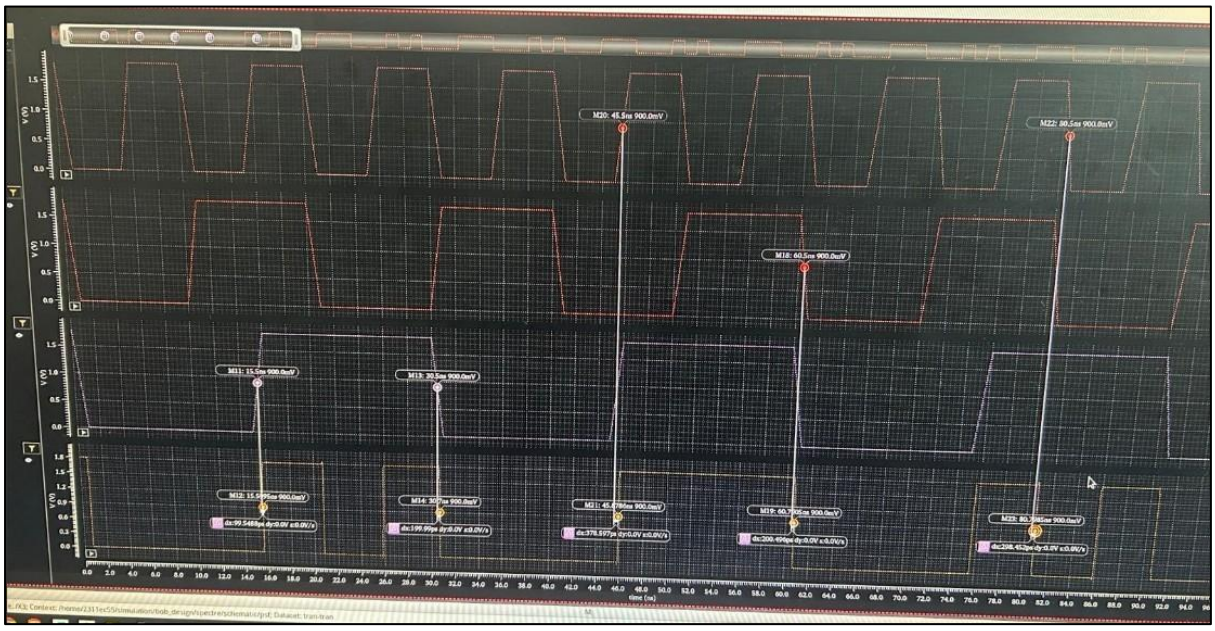


BOB DESIGN 2

PROPAGATION DELAY:

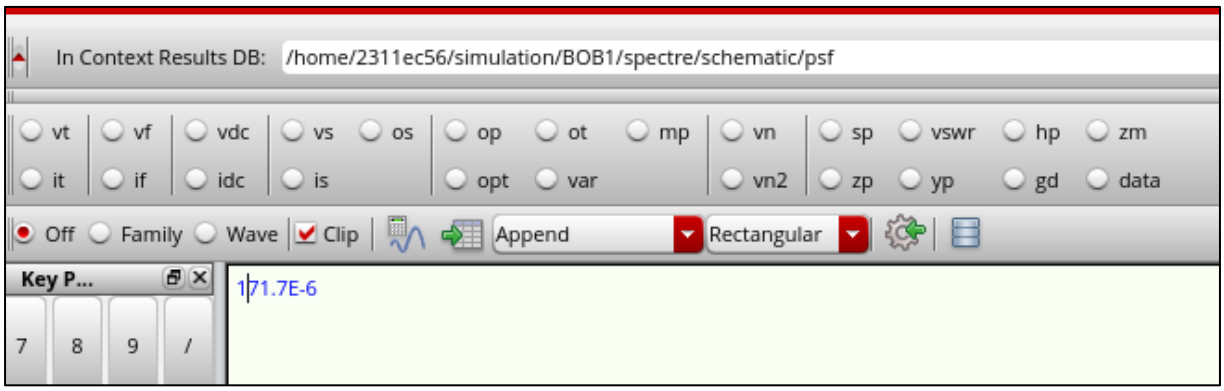


BOB DESIGN 1

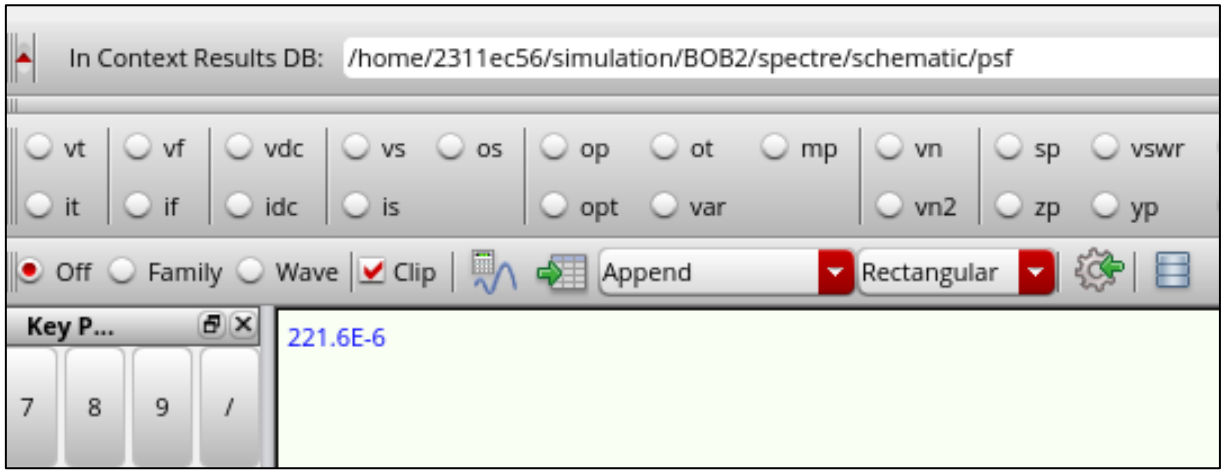


BOB DESIGN 2

POWER ANALYSIS:



BOB DESIGN 1



BOB DESIGN 2

BOB DESIGN 1:

1.Propagation Delay:

1) T_{PLH}: 0.129ns

2) T_{PHL}: 0.164ns

2.Average Power: 171.7 x 10^{-6} W.

Critical path delay (T_p): The critical path delay is the larger of the two measured delays.

T_p = 0.164 ns

Maximum switching speed (fmax) = 1/ (0.164 ns)

= 6.097 GHz

BOB DESIGN 2:

1.Propagation Delay:

1) T_{PLH}: 0.361ns

2) T_{PHL}: 0.279ns

2.Average Power: 221.6 x 10^{-6} W.

Critical path delay (T_p): The critical path delay is the larger of the two measured delays.

T_p = 0.361 ns

Maximum switching speed (fmax) = 1/ (0.361 ns)

= 2.77 GHz

Result:

Based on the quantitative analysis, Bob's Design #1 is the more optimal circuit due to its superior performance in both speed and power consumption.

- Faster Switching Speed: Design #1 has a lower worst-case propagation delay. This faster speed is a direct result of the simpler, single-gate logic used for its Z2 output.
- Lower Power Consumption: The power simulation plots clearly show that Design #1 is more power-efficient. Its circuit exhibits lower peak power spikes compared to the larger power draws seen in the plot for Design #2, indicating less energy is consumed during logic transitions.